A simple "computation"

Mishkaat :)

February 2024

1 Basic set up

Let X^h be a smooth complex submanifold of $\mathbb{CP}^m$ with it's complex dimension $dim_{\mathbb{C}} X^h = n$. Then Chow's theorem would tell us that there exists a projective variety X that can be identified with X^h. We impose the condition that the complex manifold X^h is chosen such that the projective variety X is a hypersurface i.e. a zero set of single homogeneous polynomial with the further restriction that its degree d satisfies the relation

$$d = m + 1$$

an important implication of which is that the canonical line bundle K_X of the variety X is trivial, making it a Calabi Yau manifold. So it admits a globally non vanishing top form on it as a consequence. We want to compute some "twisted" cohomology groups that will hopefully turn out to coherent sheaf cohomology and we might get GAGA type comparisons for such cohomologies.
Let $\Omega^{p,q}(X^h)$ be the space of smooth (p,q)-forms on X^h. Following [2], we deform the Dolbeault cochain complex on X^h via twisting the differential $\overline{\partial} : \Omega^{p,q}(X^h) \to \Omega^{p,q+1}(X^h)$ with a $\overline{\partial}$-closed $(0,1)$-form η as

$$\overline{\partial}_\eta \alpha := \overline{\partial}\alpha + \eta \wedge \alpha$$

for any $\alpha \in \Omega^{p,q}(X^h)$. Clearly $\overline{\partial}_\eta \circ \overline{\partial}_\eta = 0$, so we can still form a cochain complex and compute its cohomology which we denote as $H^{p,q}_\eta(X^h)$. Now consider the sheaf $\Omega^{p,q}_{X^h}$ of germs of smooth (p,q)-forms on X^h with a sheaf morphism $\overline{\partial}_\eta : \Omega^{p,q}_{X^h} \to \Omega^{p,q+1}_{X^h}$. Then the kernel sheaf of the sheaf morphism $\overline{\partial}_\eta : \Omega^{p,0}_{X^h} \to \Omega^{p,1}_{X^h}$ is defined (by [2]) to be the weight η-sheaf of holomorphic p-forms that is denoted by $\Omega^p_{X^h,\eta}$. For holomorphic functions, it is denoted as $\mathcal{O}_{X^h,\eta} := \Omega^0_{X^h,\eta}$ which we call the weight η-sheaf of holomorphic functions. The sheaf $\mathcal{O}_{X^h,\eta}$ is a locally free sheaf of $\mathcal{O}_{X^h}$-modules (where $\mathcal{O}_{X^h}$ is the structure sheaf of the ringed space $(X^h, \mathcal{O}_{X^h})$) of rank 1, because of the Grothendieck-Poincaré lemma via which the $(0,1)$-form η is locally expressible by the anti holomorphic derivative of (locally defined) smooth complex valued function. Now Oka coherence tells us that the (analytic) structure sheaf $\mathcal{O}_{X^h}$ is coherent. Thus as a locally free sheaf

of $\mathcal{O}_{X^h}$-modules, the sheaf $\mathcal{O}_{X^h,\eta}$ is coherent as well.
The analytification of the projective variety is the map

$$\lambda : X^h \to X \tag{1}$$

viewed as a map of ringed spaces. There should be a corresponding coherent sheaf $\mathcal{O}_{X,\eta}$ on the variety X, whose analytification will be the sheaf $\mathcal{O}_{X^h,\eta}$. The pullback construction of sheaves yields the following tensor product

$$\lambda^{-1}\mathcal{O}_{X,\eta} \otimes_{\lambda^{-1}\mathcal{O}_X} \mathcal{O}_{X^h} =: \mathcal{O}_{X^h,\eta} \tag{2}$$

where $\mathcal{O}_X$ is the structure sheaf on the variety X. Moreover, the weight η-sheaf of holomorphic p-forms $\Omega^p_{X^h,\eta}$ is given as the tensor product

$$\Omega^p_{X^h,\eta} = \Omega^p_{X^h} \otimes_{\mathcal{O}_{X^h}} \mathcal{O}_{X^h,\eta} \tag{3}$$

which is also locally free and coherent as well. So on the algebraic side of our story, there should be the sheaf $\Omega^p_{X,\eta}$ on the variety X for which the following

$$\lambda^{-1}\Omega^p_{X,\eta} \otimes_{\lambda^{-1}\mathcal{O}_X} \mathcal{O}_{X^h} =: \Omega^p_{X^h,\eta} \tag{4}$$

holds. Thus we have two resolutions for the sheaves $\Omega^p_{X^h,\eta}$ and $\Omega^p_{X,\eta}$

$$0 \to \Omega^p_{X^h,\eta} \to \Omega^{p,\cdot}_{X^h} \tag{5}$$

and

$$0 \to \Omega^p_{X,\eta} \to \Omega^{p,\cdot}_{X} \tag{6}$$

for which we have cohomology groups.

2 Cohomology for cohomologous η's

Following [1], we compare the cohomology groups $H^{p,q}_{\eta_1}(X^h)$, $H^{p,q}_{\eta_2}(X^h)$ for two $\overline{\partial}$-closed $(0,1)$-forms η_1, η_2 that differ by an $\overline{\partial}$-exact form

$$\eta_1 = \eta_2 + \frac{\overline{\partial} f}{f} \tag{7}$$

where f is some locally defined smooth complex valued function on some open set U of the manifold X^h, where the locality of the domain of the function f follows from the Grothendieck-Poincaré lemma. Because of the compactness of X^h, globally non vanishing holomorphic functions on X^h are constant functions. So, we will be comparing the groups $H^{p,q}_{\eta_1}(U)$, $H^{p,q}_{\eta_2}(U)$. Clearly for such η's the twisted operators are

$$\overline{\partial}_{\eta_1} := \overline{\partial} + \eta_1 \wedge \tag{8}$$

$$\overline{\partial}_{\eta_2} := \overline{\partial} + \eta_2 \wedge \tag{9}$$

which are related as

$$\overline{\partial}_{\eta_1} = \overline{\partial}_{\eta_2} + \frac{\overline{\partial} f}{f} \wedge \tag{10}$$

We now define the map

$$\Phi : H^{p,q}_{\eta_1}(U) \to H^{p,q}_{\eta_2}(U)$$

as $\Phi([\alpha]) := [f\alpha] \in H^{p,q}_{\eta_2}(U)$ For two cohomologous (p,q)-forms $\alpha_1, \alpha_2 \in H^{p,q}_{\eta_1}(U)$, they differ by $\overline{\partial}_{\eta_1}\gamma$ where $\gamma \in \Omega^{p,q-1}(U)$.
Now

$$\Phi([\alpha_1 - \alpha_2]) = [f\overline{\partial}_{\eta_1}\gamma]$$

$$\Rightarrow [f(\alpha_1 - \alpha_2)] = [(\overline{\partial}_{\eta_1})(\gamma f) + \gamma \wedge \overline{\partial} f] = [(\overline{\partial}_{\eta_2} + \frac{\overline{\partial} f}{f} \wedge)(\gamma f) + \gamma \wedge \overline{\partial} f]$$

$$\Rightarrow [f\alpha_1] - [f\alpha_2] = [\overline{\partial}_{\eta_2}(\gamma f)]$$

since the classes $[f\alpha_1]$, $[f\alpha_2]$ differ by a $\overline{\partial}_{\eta_2}$-exact form, we conclude that the class $[f\alpha] \in H^{p,q}_{\eta_2}$ and the map Φ is a well defined homomorphism.
Now we check the bijectivity of this map.
Let us pick a cohomology class $[\alpha] \in H^{p,q}_{\eta_1}$ for which $\Phi([\alpha]) = [f\alpha] = 0$. This means that the representative of this class is itself a $\overline{\partial}_{\eta_2}$-exact form. So there must be a $(p, q-1)$-form $\psi \in \Omega^{p,q-1}(U)$ such that

$$f\alpha = \overline{\partial}_{\eta_2}\psi$$

$$= \overline{\partial}\psi + \eta_2 \wedge \psi$$

$$= \overline{\partial}\psi + (\eta_1 - \frac{\overline{\partial} f}{f}) \wedge \psi$$

$$= \overline{\partial}\psi + \eta_1 \wedge \psi - \frac{\overline{\partial} f}{f} \wedge \psi$$

$$\Rightarrow \alpha = \frac{f\overline{\partial}\psi - \overline{\partial} f \wedge \psi}{f^2} + \frac{\eta_1 \wedge \psi}{f}$$

$$= \overline{\partial}(\frac{\psi}{f}) + \eta_1 \wedge (\frac{\psi}{f})$$

$$= \overline{\partial}_{\eta_1}(\frac{\psi}{f})$$

saying that the representative α is itself a $\overline{\partial}_{\eta_1}$-exact form, so the cohomology class $[\alpha] = 0$ in the group $H^{p,q}_{\eta_1}(U)$. Thus $Ker(\Phi) = 0$ and the map Φ is injective.
Now let us pick another class $[\beta] \in H^{p,q}_{\eta_2}(U)$ where the representative $\beta \in \Omega^{p,q}(U)$ is $\overline{\partial}_{\eta_2}$-closed. This means that

$$\overline{\partial}_{\eta_2}\beta = 0$$

$$\Rightarrow \overline{\partial}\beta = -\eta_2 \wedge \beta \tag{11}$$

Again from (7) we see that

$$\eta_1 \wedge \frac{\beta}{f} = \eta_2 \wedge \frac{\beta}{f} + \overline{\partial} f \wedge \frac{\beta}{f^2} \tag{12}$$

It follows that

$$\overline{\partial}_{\eta_1}(\frac{\beta}{f}) = \overline{\partial}(\frac{\beta}{f}) + \eta_1 \wedge \frac{\beta}{f}$$
$$= \frac{f\overline{\partial}\beta - \overline{\partial} f \wedge \beta}{f^2} + \eta_1 \wedge \frac{\beta}{f}$$

Using (11), (12)

$$= -\frac{\eta_2 \wedge \beta}{f} - \overline{\partial} f \wedge \frac{\beta}{f^2} + \eta_1 \wedge \frac{\beta}{f} = 0$$

which means that the $\overline{\partial}_{\eta_1}$-closed form $\frac{\beta}{f}$ represents a class in the group $H^{p,q}_{\eta_1}(U)$. Thus the arbitrarily chosen class $[\beta]$ is the image of the map Φ i.e. $\Phi([\frac{\beta}{f}]) = [\alpha]$. So the map Φ is sujective as well.
We conclude that (locally) we have the isomorphism

$$H^{p,q}_{\eta_1}(U) \simeq H^{p,q}_{\eta_2}(U) \tag{13}$$

Restricting the weight η-sheaf of holomorphic p-forms the resolution looks like

$$0 \to \Omega^p_{X^h,\eta}|_U \to \Omega^{p,\cdot}_{X^h}|_U \tag{14}$$

The isomorphism in (13) tells us that there should be a quasi isomorphism between the resolutions

$$0 \to \Omega^p_{X^h,\eta_1}|_U \to \Omega^{p,\cdot}_{X^h}|_U \tag{15}$$

$$0 \to \Omega^p_{X^h,\eta_2}|_U \to \Omega^{p,\cdot}_{X^h}|_U \tag{16}$$

3 Twisted Chern class

The typical chern classes are defined for complex vector bundles over smooth manifolds. Namely, if E is a complex vector bundle on the smooth manifold M, then the i-th chern class of the complex vector bundle E is essentially the $2i$-th cohomology group of the base manifold M i.e. $c_i(E) \in H^{2i}(M)$.
The twisted chern classes will be defined as a simple generalization of chern classes, where we consider the twisted cohomology of the base manifold. To be precise, the i-th twisted chern class is the class belonging to the $2i$-th twisted cohomology of the base manifold i.e. $c^{\psi}_i(E) \in H^{2i}_{\psi}(M)$.

4 Type-M Calabi Yau manifolds

A Calabi Yau manifold is a compact Kähler manifold with vanishing first chern class. A Calabi Yau manifold of Type-M (as we define) is a compact Kähler manifold $\tilde{M}$, whose first twisted chern class vanishes i.e. $c_1^{\psi}(\tilde{M}) = 0$. compact Kähler

References

[1] M. Islam. *Leafwise morse-novikov cohomological invariants of foliations.* PhD thesis, 2019.

[2] L. Meng. Morse-novikov cohomology on complex manifolds. *The Journal of Geometric Analysis*, 30(1):493–510, 2020.